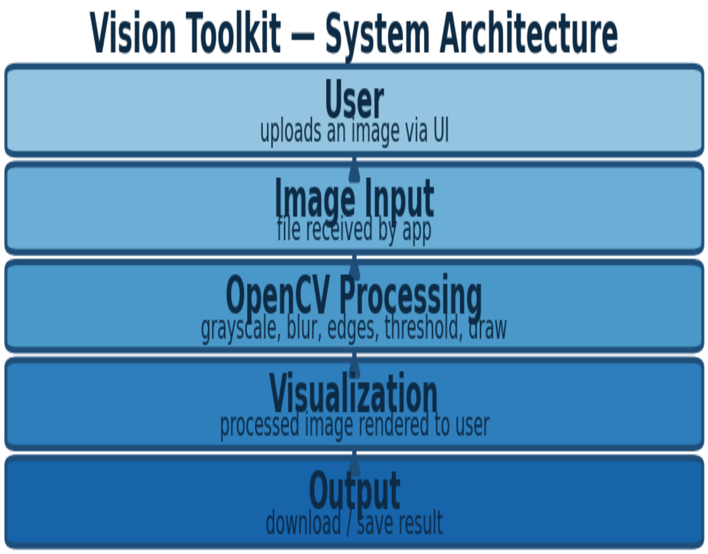
Vision Toolkit — Architecture Documentation

System design of the Week 1 application

1. Architecture Diagram



2. Stage-by-Stage Explanation

User:

The end user interacting with the application through a browser (Streamlit/web UI) or a desktop GUI window. The user's only two actions in this app are: upload an image, and choose which processing operation to apply.

Image Input:

The uploaded file is received by the application layer, decoded from its raw bytes (JPEG/PNG) into a pixel array using OpenCV's imread, and validated (file type, corrupt-file handling) before being passed further down the pipeline. This stage is the boundary between untrusted user input and the processing core — basic validation here prevents a malformed file from crashing the app.

OpenCV Processing:

The core of the application. Based on the user's selection, one or more OpenCV operations are applied to the in-memory image array:

- Grayscale conversion (cv2.cvtColor)
- Canny edge detection with adjustable thresholds (cv2.Canny)
- Gaussian / Median blur (cv2.GaussianBlur, cv2.medianBlur)
- Binary / adaptive thresholding (cv2.threshold, cv2.adaptiveThreshold)
- Drawing primitives — rectangle, circle, line, text overlay (cv2.rectangle, cv2.circle, cv2.line, cv2.putText)

Each operation is stateless and takes the current image array as input, returning a new array — this keeps the processing layer simple to test and reason about independently of the UI.

Visualization:

The processed array is re-encoded (BGR to RGB where needed for display) and rendered back to the user interface as an updated image preview, alongside metadata such as width, height, resolution, file size, and channel count.

Output:

The final processed image is made available for download/save, closing the loop from raw upload to a saved, transformed result.

3. Design Notes:

- Single responsibility per stage: input handling, processing, and visualization are kept as separate functions/modules, not one monolithic script.
- Stateless processing functions make each OpenCV operation independently unit-testable.